1986	114
1987	122
1988	129
1989	118
1990	123

16. The following table gives the number of small-scale units registered with the Directorate of Industries between 1985 and 1991. Show the growth on a trend line by the free hand method.

Year	No. of units (in'000)
1985	10
1986	22
1987	36
1988	62
1989	55
1990	40
1991	34
1992	50

17. The annual production of a commodity is given as follows :

Year	Production (in tones)
1995	155
1996	162
1997	171
1998	182
1999	158
2000	180
2001	178

Fit a straight line trend by the method of least squares.

18. Determine the equation of a straight line which best fits the following data

Year	2000	2001	2002	2003	2004
Sales (₹ '000)	35	36	79	80	40

Compute the trend values for all years from 2000 to 2004

19. The sales of a commodity in tones varied from January 2010 to December 2010 as follows:

in year 2010	Sales (in tones)
Jan	280
Feb	240
Mar	270
Apr	300
May	280
Jun	290
Jul	210
Aug	200
Sep	230
Oct	200
Nov	230
Dec	210

Fit a trend line by the method of semi-average.

20. Use the method of monthly averages to find the monthly indices for the following data of production of a commodity for the years 2002, 2003 and 2004.

2002	2003	2004
15	20	18
18	18	25
17	16	21
19	13	11
16	12	14
20	15	16
21	22	19
18	16	20
17	18	17
15	20	16
14	17	18
18	15	20

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